

Generalized Transportation Model

Until now it was considered that total supply was equal to the total demand, that is:

$$\sum_{i=1}^m a_i = \sum_{j=1}^n b_j$$

In practice, this condition is rarely verified. Nevertheless, the solution of the problem can still be obtained by the transportation method:

1. In case total supply exceeds total demand ($\sum a_i > \sum b_j$):
 - We create a dummy destination with demand equal to $\sum a_i - \sum b_j$
 - We assign “zero” to the respective “unitary transportation costs”;

- If in the optimal solution the variables corresponding to the dummy destination are different of zero, it means that these quantities don't leave the corresponding origins (!)

2. In case total supply is less than total demand ($\sum a_i < \sum b_j$):

- We create a dummy origin which supply is $\sum b_j - \sum a_i$
- We assign “zero” to the respective “unitary transportation costs”;
- If in the optimal solution the variables corresponding to the dummy origin are different of zero, it means that these quantities remain to satisfy in the respective destinations (!)

Example

Tom's bar is open only on Fridays and Saturdays at night. Normally, on those nights he sells 3 and 4 portions of a homemade liqueur, respectively.

Dick and Harry, Tom's usual suppliers, deliver at the weekend a total of 5 and 4 portions of this liquor, respectively. The prices charged depend on the day of the week and the supplier, as indicated in the following table:

	Friday	Saturday
Dick	3	2.7
Harry	2.9	2.8

(prices in MU)

Determine the most economical way for Tom to get the liquor.

Let's consider:

x_{11} = No. of liquor portions sold by Dick on Friday

x_{12} = No. of liquor portions sold by Dick on Saturday

x_{21} = No. of liquor portions sold by Harry on Friday

x_{22} = No. of liquor portions sold by Harry on Saturday

The objective is to

$$\text{Minimize } z = 3 x_{11} + 2.7 x_{12} + 2.9 x_{21} + 2.8 x_{22}$$

subject to

$$x_{11} + x_{12} \leq 5 \quad (\text{Dick sales})$$

$$x_{21} + x_{22} \leq 4 \quad (\text{Harry sales})$$

$$x_{11} + x_{21} = 3 \quad (\text{Friday purchases})$$

$$x_{12} + x_{22} = 4 \quad (\text{Saturday purchases})$$

$$x_{ij} \geq 0; i = 1, 2; j = 1, 2.$$

$$\left. \begin{array}{l} 5 + 4 = 9 \\ 3 + 4 = 7 \end{array} \right\} \rightarrow \text{We should consider a dummy destination}$$

We can obtain a first solution by the Northwest Corner method:

	Friday	Saturday	Extra	
Dick	3 3	2.7 2	0	5 2
Harry	2.9 2	2.8 2	0 2	4 2
	3	4 2	2	

$$\text{Cost} = 9 + 5.4 + 5.6 = 20 \text{ MU}$$

Occupied cells: O1D1, O1D2, O2D2, O2D3

Dual constraints (in equality form) corresponding to these cells:

$$\begin{cases} u_1 + v_1 = 3 \\ u_1 + v_2 = 2.7 \\ u_2 + v_2 = 2.8 \\ u_2 + v_3 = 0 \end{cases} \Leftrightarrow \begin{cases} u_1 = 0 \\ v_1 = 3 \\ v_2 = 2.7 \\ u_2 = 0.1 \\ v_3 = -0.1 \end{cases}$$

Empty cells: O1D3 e O2D1

The one that will be selected to be filled is cell O2D1:

	Friday	Saturday	
Dick	3 3→	2.7 → 2	...
Harry	2.9 * ←	2.8 ← 2	...

New solution:

	Friday	Saturday	Extra	
Dick	3 1 →→	2.7 →→ 4 →→	0 →→ *	5
Harry	2.9 2 ←←	2.8 ←←←←←	0 ←← 2	4
	3	4	2	

$$\text{Cost} = 3 + 10.8 + 5.8 = 19.6 \text{ MU}$$

Occupied cells: O1D1, O1D2, O2D1, O2D3

Dual constraints (in equality form) corresponding to these cells:

$$\begin{cases} u_1 + v_1 = 3 \\ u_1 + v_2 = 2.7 \\ u_2 + v_1 = 2.9 \\ u_2 + v_3 = 0 \end{cases} \Leftrightarrow \begin{cases} u_1 = 0 \\ v_1 = 3 \\ v_2 = 2.7 \\ u_2 = -0.1 \\ v_3 = 0.1 \end{cases}$$

Empty cells: O1D3 e O2D2

The one that will be selected to be filled is cell O1D3.

New solution:

	Friday	Saturday	Extra	
Dick	3	2.7	0	5
Harry	2.9	2.8	0	4
	3	4	2	

(Verify that the obtained solution is optimal)

The optimal solution will be:

On Friday, buy 3 portions of liquor to Harry

On Saturday, by 4 portions of liquor to Dick

Minimum cost = 19.5 MU